FORMULA 1 RACING INTELLIGENCE

Project Overview and Dataset Description

1. Problem Statement and Project Objectives

1.1 Problem Statement

Formula 1 is a competitive motorsport in which driver performance, team strategy, qualifying positions, lap times, pit stops, tyre selection, weather conditions, and circuit characteristics influence race outcomes. Understanding these factors requires the analysis of large amounts of racing data.

Raw datasets often contain numerous records and variables, making it difficult to identify meaningful patterns through tables alone. Data visualization provides a way to convert numerical information into clear and understandable insights.

The Formula 1 Racing Intelligence project focuses on analyzing a provided Formula 1 dataset using Tableau. The project involves creating charts and visualizations, developing a dashboard, and presenting the results through a website deployed using Render.

1.2 Project Objectives

The main objectives of this project are:

1. To explore and understand the structure of the provided Formula 1 dataset.
2. To analyze race results, driver performance, qualifying results, lap statistics, championship standings, and circuit information.
3. To identify patterns and trends using statistical analysis.
4. To create visualizations, including bar charts, pie charts, and other suitable charts in Tableau.
5. To combine selected visualizations into a dashboard.
6. To communicate important findings through graphical representations.
7. To deploy a project website using Render.
8. To document the project workflow, findings, and conclusions.

2. Data Collection and Dataset Description

2.1 Data Collection

The project uses an Excel workbook named *F1 Racing Project Dataset.xlsx*, which was provided for the Formula 1 Racing Intelligence project.

The workbook contains five worksheets covering different aspects of Formula 1 racing. These include race results, lap intelligence, qualifying performance, championship standings, and circuit information.

The dataset serves as the primary source for the statistical analysis and visualizations created in Tableau.

2.2 Dataset Overview

The workbook contains a total of 13,748 records across five worksheets.

Worksheet	Records	Description
race_results	6,464	Race outcomes, points, pit stops, weather, and overtaking statistics
lap_intelligence	3,814	Lap times, sector times, tyre compounds, DRS, and speed measurements
qualifying_results	3,130	Qualifying positions, lap times, and speed measurements
championship_standings	311	Driver standings, points, wins, and podiums
circuits	29	Circuit names, locations, lengths, and track characteristics
Total	13,748	Records across all five worksheets

The dataset covers seasons from 2010 to 2026.

The race_results worksheet contains 334 distinct race IDs, 19 distinct driver names, 10 teams, and 24 distinct circuit IDs.

2.3 Important Dataset Variables

The dataset includes several variables that can be used for analysis:

- Driver and team names for comparing performance.
- Season and race IDs for organizing records.
- Finishing positions and grid positions for examining race outcomes.
- Points and laps led for comparing driver performance.
- Pit stops and pit-stop times for exploring race strategy.
- Lap times and sector times for examining lap performance.
- Tyre compounds and tyre age for analyzing tyre-related data.

- Qualifying positions and gaps to pole for comparing qualifying performance.
- Weather conditions and circuit characteristics for exploring race environments.

These variables provide different perspectives on Formula 1 racing and support the development of meaningful visualizations.